

Practice Quiz 03B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 03

Nested quantifiers and negating quantified statements.

1. Consider

$$(i) \forall x \in \mathbf{R} \exists y \in \mathbf{R} (x + y = 1) \quad (ii) \exists y \in \mathbf{R} \forall x \in \mathbf{R} (x + y = 1).$$

(a) State each in plain English and say whether it is true or false.

Solution Statement (i) says that for every real x there is a real y whose sum with x is 1. It is true: take $y = 1 - x$. Statement (ii) says that one fixed real y has sum 1 with every real x . It is false. ■

(b) Negate (ii), with no negation symbol to the left of a quantifier.

Solution Its negation is

$$\forall y \in \mathbf{R} \exists x \in \mathbf{R} (x + y \neq 1).$$

■